

Education

Federal University of Paraíba

2026 – present

MSc. in Physics. Advisor: Dionísio Bazeia 

Federal University of Campina Grande

2021 – 2025

BSc. in Physics. Ranked in the top 10% of the class.

Thesis: Effects of photon-subtraction and photon-addition in the Schrödinger cat state.

Publications

Preprints

Jhordan Santiago and Petr Steindl. Statistics and sub-Planck structures of photon-added crystallized cat states (to appear).

Jhordan Santiago. Higher-order nonclassicality criteria for photon-subtracted and photon-added states via the normalization constant, arXiv: [2511.21801](https://arxiv.org/abs/2511.21801).

Published

Jhordan Santiago and Petr Steindl . Nonclassicality of multiphoton-added cat states, *Physical Review A*, **113**, L041701, arXiv: [2601.08894](https://arxiv.org/abs/2601.08894).

Jhordan Santiago. Photon statistics for Fock and coherent states interfering in a beamsplitter, *Revista Brasileira de Ensino de Física*, **47**, 2025.

Conference Presentations

40th North and Northeast Physics Meeting 2025

Poster on “Quantum interference between photon-added coherent states”, Danievertton Moretti (presenting) and Jhordan Santiago.

11th Week of Physics – Universidade Federal de Campina Grande 2025

Poster on “Inseminating the cat: effects of photon-addition and subtraction on the Schrödinger cat state”, Jhordan Santiago (presenting) and Danievertton Moretti.

Summer Program in Physics – Federal University of Pernambuco 2025

Poster and Oral Communication on “Basic statistics in addition and subtraction processes of a single-photon using Fock and coherent states”, Jhordan Santiago (presenting) and Danievertton Moretti.

39th North and Northeast Physics Meeting 2024

Poster on “Addition and subtraction of single-photons in the Mach-Zehnder interferometer: a brief study of the Wigner functions”, Jhordan Santiago (presenting) and Danievertton Moretti.

1st Workshop on Black Holes and Gravitational Analogues 2024

Poster on “Quantum mechanics in phase space and the Wigner function”, Jhordan Santiago (presenting) and Danievertton Moretti.

BINGO Collaboration Meeting 2023, 38th North and Northeast Physics Meeting 2023

Poster on “Quantum interference at a beamsplitter: study of the Wigner function, click probability, and average photon counting with an input state of the type $|n\rangle|\alpha\rangle$ ”, Jhordan Santiago (presenting) and Danievertton Moretti.

10th Week of Physics – Universidade Federal de Campina Grande 2023

Poster on “Photon statistics at a beamsplitter”, Jhordan Santiago (presenting) and Danievertton Moretti.

Winter Program in Physics – University of São Paulo 2019

Oral Communication on “Blackbody radiation”, Jhordan Santiago (presenting).

Science Outreach

[Drops of Physics](#)

2022 – 2025

It aims to offer to middle and high school teachers an experimental teaching alternative that promotes discussion of physical phenomena. We regularly went to public schools and present a collection of interactive experiments for students. Funded by Brazil’s state funding agency CNPQ.

In Traffic, Physics is the one who beeps

2022 – 2022

Developed an eletronic prototype for measuring reaction time based on a stimulus.

[Universo Racionalista](#)

2019 – 2024

Editor and writer of science news to the former biggest SciCom website in Portuguese.

Skills

Languages

Python, $\text{\LaTeX}$ and AI tools.

Portuguese (native), English (fluent), Spanish (intermediate).